

Class Notes

Best Response and Learning in Games

It would be desirable that natural game playing strategies quickly lead players to either find the equilibrium or at least converge to an equilibrium in the limit. Maybe the most natural “game playing” strategy is that of following “best response”.

Consider an action profile a , and a player i . Using the action profile, a player i gets the value or utility $u_i(a_i, a_{-i})$. Changing the strategy a_i to some other strategy a'_i , the player can change his utility to $u_i(a'_i, a_{-i})$, assuming that all other players stick to their strategies in a_{-i} . We say that a change from strategy a_i to a'_i is an **improving response** for player i if $u_i(a'_i, a_{-i}) > u_i(a_i, a_{-i})$ and a best response if a'_i maximizes the player's utility. Playing a game by repeatedly allowing some player to make an improving or a best response move is perhaps the most natural game play.

In some games, such as the Prisoner's Dilemma or the Coordination Game, this dynamic leads the players to a Nash equilibrium in a few steps. In other games the players will not reach the equilibrium in a finite number of steps, but the action profile will converge to the equilibrium.

In other games, the play may cycle, and not converge. A simple example is matching pennies, where the players will cycle through the 4 possible action profiles if they alternate making best response moves. While this game play does not find a pure equilibrium (as none exists) in some sense we can still say that best response converges to the equilibrium: the average payoff for the two players converges to 0, which is the average payoff; and even the frequencies at which the 4 possible strategy vectors are played converge to the probabilities in equilibrium (1/4 each).

Example game

		player 2		
		A	B	C
player 1	A	5, 3	1, 4	3, 3
	B	4, 2	3, 3	2, 1
	C	2, 2	2, 4	1, 3

In red best responses for player 1, in blue best responses for player 2, the only Nash is B,B. Notice that A,C would have the same payoff but is not a Nash equilibrium and A,A is no worse but is not again is not a Nash equilibrium.

Show that iteration from a random point converges to B,B in few steps.

In next section we will also see that for player 1, action B dominates action C while for player 2 action B dominates both action A and action C. Computing both best responses and dominated actions player 2 can know that player 1 will never chose C but she doesn't care because the best strategy is B independently on what player 1 does. With the same reasoning player 1 will know that with certainty player 2 will chose B thus it choses its best response accordingly.

Dominated actions

You drive up to a red traffic light. The left lane is free; in the right lane there is a car that may turn right when the light changes to green, in which case it will have to wait for a pedestrian to cross the side street. Assuming you wish to progress as quickly as possible, the action of pulling up in the left lane “strictly dominates” that of pulling up in the right lane. If the car in the right lane turns right then you are much better off in the left lane, where your progress will not be impeded; and

even if the car in the right lane does not turn right, you are still better off in the left lane, rather than behind the other car.

In any game, a player's action "strictly dominates" another action if it is superior, no matter what the other players do. Formally: in a strategic game with ordinal preferences, player i 's action a_i'' **strictly dominates** her action a_i' if

$$u_i(a_i'', a_{-i}) > u_i(a_i', a_{-i}) \quad \forall a_{-i} \text{ of the other players}$$

The concept of domination can be relaxed accepting that the action a_i'' is not superior but at least as good as a_i' . In this case we have that in a strategic game with ordinal preferences, player i 's action a_i'' **weakly dominates** her action a_i' if

$$u_i(a_i'', a_{-i}) \geq u_i(a_i', a_{-i}) \quad \forall a_{-i} \text{ of the other players}$$

and

$$u_i(a_i'', a_{-i}) > u_i(a_i', a_{-i}) \quad \text{for some } a_{-i} \text{ of the other players}$$

Dominated actions have the following useful properties that help to find Nash equilibria

- A strictly dominated action is not a best response to any actions of the other players: whatever the other players do, some other action is better
 - A strictly dominated action is not used in any Nash equilibrium.
- A strictly dominated action is incompatible not only with a steady state, but also with rational behavior by a player who confronts a game for the first time.
- In a strict Nash equilibrium (no player's equilibrium action is weakly dominated)

EXAMPLE: in the prisoner dilemma the free ride option of finking dominates the quiet action independently on the other player's action.

		Suspect 2	
		Quiet	Fink
Suspect 1	Quiet	2,2	0,3
	Fink	3,0	1,1

In fact, for a player the utility of Fink is always higher than that of Quiet independently on the choice of the other player. E.g. if suspect 2 decides for quiet fink is $3 > \text{quiet } 2$. If suspect 2 decides for fink, the utility of fink is $1 > \text{quiet } 0$.

EXAMPLE: public good game. We provide here a simplified version of the public good game with only two players. The same considerations hold for many players. Two people have 1\$ and can contribute to a public fund. The total collected is multiplied for a constant (i.e., 1.5). Then the fund is equally shared between the two players.

		player 2	
		contribute	Don't contribute
player 1	contribute	1.5, 1.5	0.75, 1.75
	Don't contribute	1.75, 0.75	1, 1

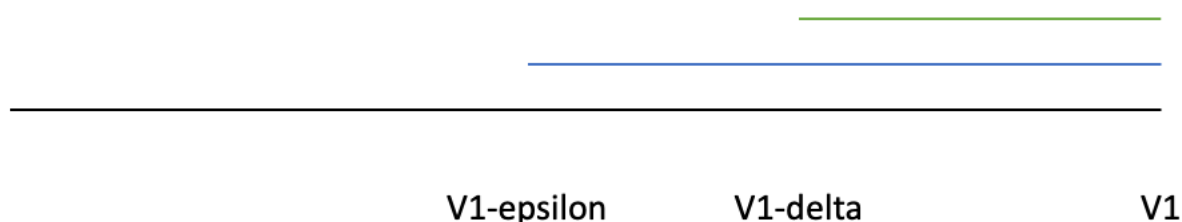
The only Nash equilibrium is “don’t contribute” for both players. This action dominates contribute because regardless what the other player does, don’t contribute is always more convenient. However, the action (contribute, contribute) would be better for both. This game introduces the concept of free rider: namely a player that making its best choice obtains a worse outcome. These games show the need of external incentives to make contributing more convenient.

Vickrey Auction: Designing Games with Dominant Strategy Solutions

Perhaps the most common situation in which we need to design a game is an auction. Suppose that we are faced with designing an auction to sell a valuable painting. To model this situation as a game, assume that each player (bidder) i has a value v_i for the painting. His value or payoff for not winning it is 0, and his payoff for winning it at a price of p is $v_i - p$. The strategy of each player is simply his bid. What is a good mechanism (or game) for selling this painting? Here we are considering single-shot games, so assume that each player is asked to state his bid for the painting in a sealed envelope, and we will decide who to award the painting to and for what price, based on the bids in the envelopes.

Perhaps the most straightforward auction would be to award the painting to the highest bidder and charge him his bid. This **game does not have a dominant strategy solution**. A player's best strategy (bid) depends on what he knows or assumes about the strategies of the other players. Deciding what value to bid seems like a hard problem, and may result in unpredictable behavior.

To show why the game does not have a dominant strategy consider only two players: p_1 with value v_1 and p_2 with value $v_2 < v_1$. If player 1 offers v_1 is payoff is 0, thus she should offer less but more than what player 2 offers, otherwise her payoff would be 0 again. While it would seem that v_1 -epsilon dominates $v_1 - \delta$ (with $\delta < \epsilon$), this is not true regardless what p_2 does, but only if p_2 's offer is lower than v_1 -epsilon, otherwise v_2 wins the auction and $v_1 - \epsilon$ has payoff 0 but $v_1 - \delta$ has payoff $\delta > 0$.



Due to the difficulty of defining a strategy for this auction let's see what happens if we do not make it in one shot, but we convert it into an iterative process. An English auction is a process of selling where price starts low and increases as buyers bid for the item until one buyer is left willing to pay a certain amount and a higher bid isn't received. For player 1 to win it is enough to offer just above the second higher player. When p_1 and p_2 compete, p_2 will stop offering v_2 as it is her highest bid (payoff becomes 0) thus p_1 can offer $p_2 + \epsilon$ winning and achieving a payoff of $v_1 - v_2 - \epsilon$. Being ϵ very close to 0 we can argue that p_1 pays the second price.

Second-price sealed-bid auctions can be modelled as this game

Players: The n bidders, where $n \geq 2$.

Actions: The set of actions of each player is the set of possible bids (nonnegative numbers).

Preferences: The payoff of any player i is $v_i - b_j$, where b_j is the highest bid submitted by a player other than i if either b_i is higher than every other bid, or b_i is at least as high as every other bid and the number of every other player who bids b_i is greater than i . Otherwise player i 's payoff is 0.

The second price auction has the remarkable property that **each player's dominant strategy is to report his true value as bid, independently of the strategies of the rest of the players!**

Observations:

- Since payoff is $v_i - b_i$ no one would offer more than her true value because otherwise other players behavior could lead to a negative payoff (I win and the second highest offer is higher than v_i).
- If $v_i \neq \max_{j \in [1, n]} v_j$ the player won't win thus her payoff will be 0 regardless what she makes. Offering less would only reduce her chances to win. In this sense offering her true value dominates the other actions since in terms of expected payoff the probability of payoff $\neq 0$ is maximum (i.e. the probability of being the winner)
- If v_i is the maximum value, offering less than v_i does not change the payoff unless it becomes the second highest bid (causing the payoff to drop to 0). Again, in terms of expected payoff v_i maximizes the probability of being the winner.

Let's see the Nash equilibria

Case 1: $(b_1, \dots, b_n) = (v_1, \dots, v_n)$: each player's bid is equal to her valuation of the object. Because $v_1 > v_2 > \dots > v_n$, the outcome is that player 1 obtains the object at the price b_2 ; her payoff is $v_1 - b_2$ and every other player's payoff is zero. This profile is a Nash equilibrium by the following argument.

- If player 1 changes her bid to some other price at least equal to b_2 then the outcome does not change (recall that she pays the second highest bid, not the highest bid). If she changes her bid to a price less than b_2 then she loses and obtains the payoff of zero.
- If some other player lowers her bid or raises it to some price at most equal to b_1 then she remains a loser; if she raises her bid above b_1 then she wins but, in paying the price b_1 , makes a loss (because her valuation is less than b_1).

Case 2: $(b_1, \dots, b_n) = (v_1, 0, \dots, 0)$. In this equilibrium, player 1 obtains the object and pays the price of zero

- As before player 1 has no interest in lowering the offer as she still pays 0
- Losers have no interest in increasing their offer as they either remain losers or win paying b_1 and having a negative payoff

Case 3: $(v_2, v_1, 0, \dots, 0)$, in which player 2 obtains the object at the price v_2 and every player (including player 2) receives the payoff of zero. This action profile is a Nash equilibrium by the following argument.

- If player 1 raises her bid to v_1 or more, she wins the object but her payoff remains zero (she pays the price v_1 , bid by player 2). Any other change in her bid has no effect on the outcome.

- If player 2 changes her bid to some other price greater than v_2 , the outcome does not change. If she changes her bid to v_2 or less she loses, and her payoff remains zero.
- If any other player raises her bid to at most v_1 , the outcome does not change. If she raises her bid above v_1 then she wins, but in paying the price v_1 (bid by player 2) she obtains a negative payoff.

Observe that the result is not surprising since in an English auction every new bid is an improving action.